

0.反射前置

$$\omega_n = \begin{cases} \min\{\alpha \mid \alpha \in Ord \wedge \|\alpha\| = \aleph_n\} & n > 0 \\ 1 & n = 0 \end{cases}$$

对于反射定义等全体序数在当前情况下, 均指 $\in \omega_2$

以 $\{\alpha \mid \alpha < \omega_1\}$ 作为 ω_1 的基本列

$$\Pi_0 = Ord$$

$$\forall a, b \in Refl_n \rightarrow a \cap b \in Refl_n \wedge \forall \alpha (\alpha \in Ord \wedge \alpha \geq n) \rightarrow \Pi_\alpha \text{ onto } A, \Pi_\alpha \text{ onto } B \in Refl_n$$

$$\Omega_n = \begin{cases} n \text{ th } \Pi_2 & \exists \beta (\alpha = \beta + 1) \\ \beta \text{ th } \Pi_1 \text{ onto } \Pi_2 & \exists \beta (\alpha = \omega * \beta) \end{cases}$$

$$Inc(\alpha) = \begin{cases} Inc(\mu) + Inc(\nu) & \alpha = \mu + \nu \\ \psi_{\omega_1 + Inc(\nu)}(Inc(\mu)) & \alpha = \psi_\nu(\mu) \wedge \nu > 0 \\ n & \alpha = n \wedge n \in \mathbb{N}_0 \\ (\omega_1 + n) \text{ th } A & A \in Refl_2 \end{cases}$$

$$(A \text{ onto})^\alpha B = \begin{cases} A \text{ onto } (A \text{ onto})^\beta B & \alpha = \beta + 1 \\ \{\alpha \mid \forall \nu < \alpha, \exists \beta \in (A \text{ onto})^\nu B\} & cf(\alpha) = \omega \\ \{\nu \mid \forall \beta < \nu, \nu \in (A \text{ onto})^{\alpha[\beta]} B\} & cf(\alpha) = \omega_1 \end{cases}$$

2. BOCF

$$C(\alpha, \beta, 0) = \Omega_\beta \cup \{\omega_1\} \cup \{\Pi_n \mid 0 < n < \omega\}$$

$$C(\alpha, \beta, n+1) = \{\nu + \mu \mid \nu, \mu \in C(\alpha, \beta, n)\} \cup \{\psi_\nu(\mu) \mid \nu, \mu \in C(\alpha, \beta, n) \wedge \mu \in$$

$$C(\mu, \nu) \wedge ((\beta < \omega_1 \wedge \nu \geq \omega_1 \wedge \mu < Inc(\alpha)) \vee (\mu < \alpha))\} \cup \{\mu \text{ th } A \mid A \in Refl_2 \cap$$

$$C(\alpha, \beta, n) \wedge \mu \in C(\alpha, \beta, n) \wedge ((\mu \geq \omega_1 \wedge \mu < Inc(\alpha)) \vee (\mu < \alpha))\} \cup \{(A \text{ onto})^\mu B \mid$$

$$\mu \in C(\alpha, \beta, n) \wedge A, B \in C(\alpha, \beta, n) \wedge B \in Refl_2 \wedge ((\mu \geq \omega_1 \wedge \mu < Inc(\alpha)) \vee (\mu < \alpha))\} \cup C(\alpha, \beta, n)$$

$$C(\alpha, \beta) = \bigcup_{n < \omega} C(\alpha, \beta, n)$$

$$\psi_\beta(\beta) = \min\{\mu \mid \mu \in Ord \wedge \mu \notin C(\alpha, \beta)\}$$